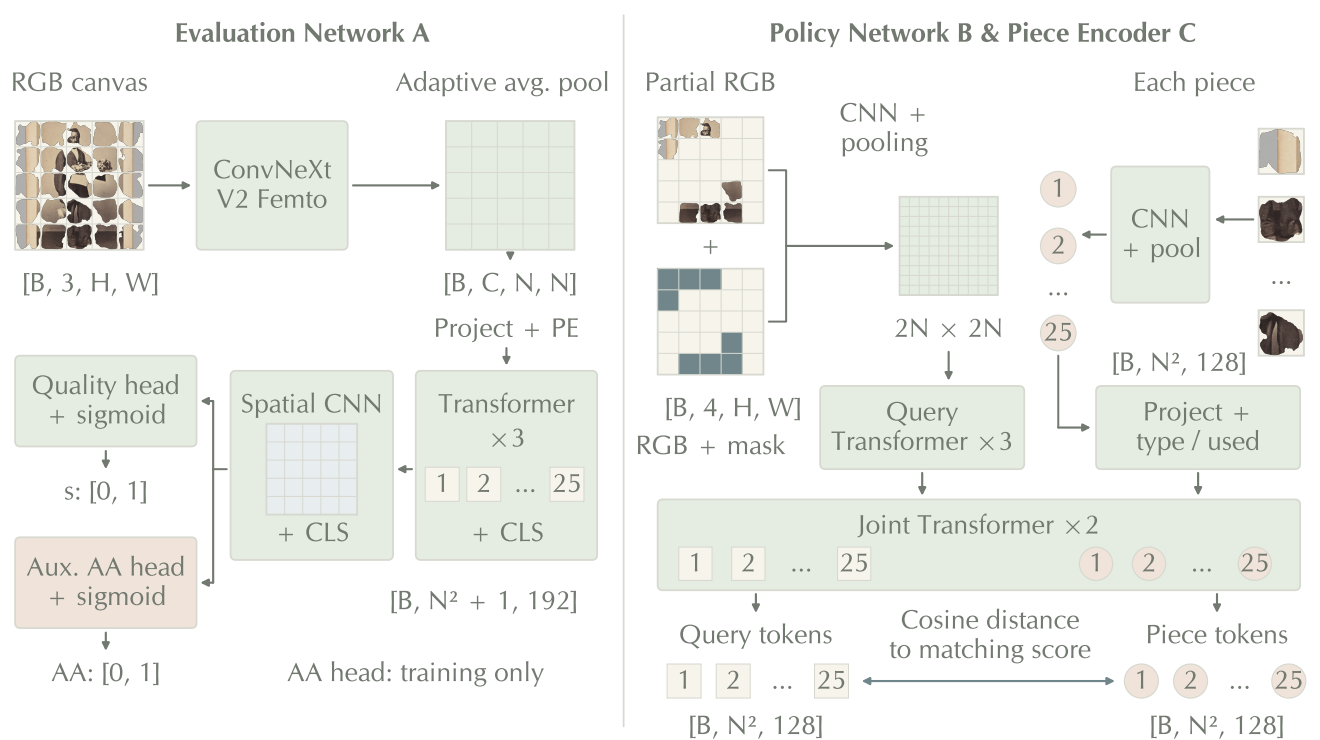

Evaluation Network A
RGB canvas
Adaptive avg. pool
ConvNeXt V2 Femto
[B, 3, H, W]
[B, C, N, N]
Project + PE
Quality head + sigmoid
s: [0, 1]
Spatial CNN
+ CLS
Transformer ×3
1 2 ... 25
+ CLS
Aux. AA head + sigmoid
AA: [0, 1]
[B, N² + 1, 192]
AA head: training only
Policy Network B & Piece Encoder C
Partial RGB
CNN + pooling
+
[B, 4, H, W]
RGB + mask
2N × 2N
Each piece
CNN + pool
[B, N², 128]
Query Transformer ×3
Project + type / used
Joint Transformer ×2
1 2 ... 25
1 2 ... 25
Query tokens
Cosine distance to matching score
Piece tokens
1 2 ... 25
[B, N², 128]
1 2 ... 25
[B, N², 128]